

Security flaw

According to a recent survey conducted by Harris Interactive, about 12 per cent of IT professionals admitted to knowingly violating IT policy in order to get work done

Another boy wonder?

Sunny Vaghela, 22 year old self proclaimed cyber security expert is planning to launch a website to analyse cyber crime complaints. It will be run by a team of ethical hackers who are students.

Work@tech

strong fundamentals in network communication, and very good analytical skills. Goyal had a fair warning to all aspirants of this field “Read read and read”. Unlike other domains in the IT sector, in information security you constantly need to be updated on things. “Engineers finish their B.E and then they become complacent,” he adds. This will certainly not do when it comes to IT security. Apart from these, good IQ, analytical skills and a natural curiosity about how networks work cannot be ignored.

Job roles

IT security or information security as a field has a very broad coverage. Opportunities exist in managing/administrating security devices, reporting/auditing on IT infrastructure, penetration testing, Compliance reporting and management, as well as various engineering/security research roles in security product companies like Symantec who build security software products. You can specialise in one aspect or you can have overall coverage of all the domains.

The responsibilities will vary depending on the domain which the security specialist belongs to. Most large organisations have segregation of duties defined so security personnel can focus on those specific domains. In the Security Response domain, AV analysts focus on providing quick mitigation to security threats which are observed in the network. Experts are responsible for providing mitigation to vulnerabilities as they are disclosed and providing mitigation for security risks whose information can be derived from security patches released by vendors.

They get to work on products that secure the email infrastructure or the instant messaging infrastructure of an organisation by keeping out spam, spim and viruses.

“Security domain encompasses

Certification reckoner

CISA	Certified Information Systems Auditor	Audit oriented	Information Systems Audit and Control Association www.isaca.org . Exams conducted in India
GCIA	Certified Intrusion Analyst	Technology and networking oriented	www.giac.org Online exam
GCIH	Certified Incident Handler	Technology oriented	www.giac.org Online exam
CISSP	Certified Information Systems Security Professional	More technology oriented	International Information Systems Security Certification Consortium www.isc2.org . Exams conducted in India
CEH	Certified Ethical Hacker	Very basic course and exam	Conducted by European Council

multiple sub areas of specialisation like anti-fraud protection, anti-spam protection, network intrusion protection, desktop protection, compliance management, data leakage prevention, incident response”, informs Shantanu Ghosh

The most glamorous sub-domain however is ethical hacking or in security parlance – penetration testing. We spoke to a penetration testing professional, Siddhesh Shenvi working for AAA Technologies, which specialises in the field of Information Systems (IS) Audit and security consulting. “We have to sign NDAs (Non Disclosure Agreements) and operate under full authority given by our clients to penetrate and test the vulnerability of their systems,” says Siddhesh. He added,

“Every industry which has data at rest or data in motion needs to have IT Security personnel”

“This being the most challenging domain, we’re working with the newest technologies in the sector. Suppose a system is behind a firewall, put in DMZ mode, or there are access control lists, and dual authentication systems;

we bypass these security measures and gain access to the system”. He also informed us that despite being the most challenging, penetration testing is by far the most interesting aspect of IT security. “Here learning is a continuous, never ending process. Whatever knowledge you acquire is a treat, and still just a drop in the ocean”.

Career paths

The career path is dependent on the work domain the individual is currently in. “At Security Response Labs, engineers can grow from Security Engineers/Analysts, to Principal Engineers/Analysts,” says Ghosh from Symantec. As they grow in these roles they will address standalone malware threats as well as network security issues.

Software development professionals

Sub-domains under IT security

Information Security Audit
Application Audit
Business Continuity Management
Fraud/Forensics Investigation
IT security consulting
Research
IS security management for business
Network penetration testing
Database/Network Administration
Network hardening
Policy formulation

are expected to have thorough knowledge of computer networking, various programming languages and the flaws that may exist while programming in these languages.

Last words

As per a survey conducted by (ISC)2, the unfavourable economic situation did not negatively impact the security while on the contrary 51.4 per cent in the Asia Pacific region received salary increases in 2009. This career presents several international opportunities in the Asia Pacific region as well as the western world. This is because the shortage of IS professionals abroad is even more acute. Besides, the public sector in India is also a great source of employment. 



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